Luiz Felipe Vecchietti

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Last Updated: October, 2026

Research Interests

My research aims to develop safe artificial intelligence algorithms for multi-agent systems that advance our knowledge of how humans and machines learn, interact, and collaborate. In particular, I am interested in multi-agent systems (collaborative and collaborative-competitive environments), robotics (end-to-end control, safety risks, and hardware security), and reinforcement learning (continual learning, credit assignment, multi-agent RL). My research is being applied to diverse domains such as robotics, natural language processing, and AI for science.

Academic Positions

Max Planck Institute for Security and Privacy (MPI-SP) *Bochum, Germany, Nov. 2024 - Present*

Postdoctoral Researcher - Data Science for Humanity Group. Advisor: Meeyoung Cha.

Institute for Basic Science (IBS) *Daejeon, South Korea, Oct. 2021 - Sep. 2024*

Postdoctoral Researcher - Data Science Group. Advisor: Meeyoung Cha.

Mechanical Engineering Research Institute, KAIST *Daejeon, South Korea, Mar. 2021 - Sep. 2021*

Postdoctoral Researcher.

Education

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea, Aug. 2017 - Feb. 2021*

Ph.D. in Green Transportation. Advisor: Dongsoo Har.

Thesis topic: Performance Enhancement in Multigoal Reinforcement Learning using Hindsight Experience Replay.

COPPE - Federal University of Rio de Janeiro (UFRJ) *Rio de Janeiro, Brazil, Mar. 2015 - Apr. 2017*

M.Sc. in Electrical Engineering. Advisor: Fernando Gil Vianna Resende Junior.

Thesis topic: Comparison between rule-based and data-driven Natural Language Processing algorithms for Brazilian Portuguese Speech Synthesis.

Federal University of Rio de Janeiro (UFRJ) *Rio de Janeiro, Brazil, Feb. 2009 - Dec. 2014*

B.Sc. in Electronics and Computer Engineering.

Korea Advanced Institute of Science and Technology (KAIST) *Daejeon, South Korea, Feb. 2013 - Feb. 2014*

Exchange Student in Electrical Engineering (Science without Borders Program).

Publications

[1] **L. F. Vecchietti**, B. N. Wijaya, W. Dong, M. Cha, "**Self-Evolving Agents Should Build Internal and External Models of the World**," in NeurIPS, 2026.

[2] M. S. Locatelli, F. Tonucci, J. Kwon, **L. F. Vecchietti**, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha, "**Textual Supervision Enhances Geospatial Representations in Vision-Language Models**," in International Conference on Machine Learning (ICML), 2026.

[3] N. Koeppe, **L. F. Vecchietti**, D. Han, D. Li, S. W. Lee, "**Mitigating Plasticity Loss through Architectural Design in Continual Learning**," in International Conference on Machine Learning (ICML), 2026.

[4] J. Kim, J. Kwon, **L. F. Vecchietti**, W. Dong, J. Kim, M. Cha, "**Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions**," in ACL Findings, 2026.

[5] **L. F. Vecchietti**, M. Lee, B. Hangeldiyev, B. N. Wijaya, H. Jung, H. Park, T-K Kim, M. Cha, H. M. Kim, "**Interpretable Machine Learning for Protein Science: Structure, Function, and Interactions**," in ACM Computing Surveys, 2026.

[6] J. Kwon, **L. F. Vecchietti**, S. Park, M. Cha, "**Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment**," in AAAI, AI Alignment Track, 2026.

[7] **L. F. Vecchietti**, B. N. Wijaya, et al., "**Artificial intelligence-driven computational methods for antibody design and optimization**," mAbs, 2025, doi: 10.1080/19420862.2025.2528902.

[8] J. Yun, S. Yang, J. H. Kwon, **L. F. Vecchietti**, M. Cha, J. E. Oh, H. M. Kim, "**Computational Design and Glycoengineering of Interferon-Lambda for Nasal Prophylaxis against Respiratory Viruses**," Advanced Science, 2025.

[9] M. Lee, **L. F. Vecchietti** *(co-first author)*, H. Jung, H. J. Ro, M. Cha, H. M. Kim, "**Robust Optimization in Protein Fitness Landscapes Using Reinforcement Learning in Latent Space**," in International Conference on Machine Learning (ICML), 2024 *(Spotlight Poster)*.

[10] M. Seo, **L. F. Vecchietti**, S. Lee, D. Har, "**Rewards Prediction-Based Credit Assignment for Reinforcement Learning With Sparse Binary Rewards**," in IEEE Access, vol. 7, pp. 118776-118791, 2019, doi: 10.1109/ACCESS.2019.2936863.

[11] T. B. Ribeiro, **L. F. Vecchietti**, et al., "**Overabundance of abelisaurid teeth in the Acu Formation (Albian-Cenomanian), Potiguar Basin, Northeastern Brazil: morphometric, cladistic and machine learning approaches**," Journal of Vertebrate Paleontology, 2025.

[12] S. Mishra, P. K. Rajendran, **L. F. Vecchietti**, D. Har, "**Sensing accident-prone features in urban scenes for proactive driving and accident prevention**", 2023, IEEE Transactions on Intelligent Transportation Systems.

[13] **L. F. Vecchietti**, M. Seo, D. Har, "**Sampling Rate Decay in Hindsight Experience Replay for Robot Control**," in IEEE Transactions on Cybernetics, 2020, doi: 10.1109/TCYB.2020.2990722.

[14] **L. F. Vecchietti**, T. Kim, K. Choi, J. Hong, D. Har, "**Batch Prioritization in Multigoal Reinforcement Learning**," in IEEE Access, vol. 8, pp. 137449-137461, 2020, doi: 10.1109/ACCESS.2020.3012204.

[15] C. Hong, I. Jeong, **L. F. Vecchietti**, D. Har, J. -H. Kim, "**AI World Cup: Robot Soccer-Based Competitions**," in IEEE Transactions on Games, 2021, doi: 10.1109/TG.2021.3065410.

[16] T. Kim, **L. F. Vecchietti**, K. Choi, S. Sariel, D. Har, "**Two-stage training algorithm for AI robot soccer**", in PeerJ Computer Science, 7:e718, 2021, https://doi.org/10.7717/peerj-cs.718.

[17] S. Lee, **L. F. Vecchietti**, H. Jin, J. Hong, D. Har, "**Power Management by LSTM Network for Nanogrids**," in IEEE Access, vol. 8, pp. 24081-24097, 2020, doi: 10.1109/ACCESS.2020.2969460.

[18] S. Lee, H. Jin, **L. F. Vecchietti**, J. Hong, D. Har, "**Short-Term Predictive Power Management of PV-Powered Nanogrids**," in IEEE Access, vol. 8, pp. 147839-147857, 2020, doi: 10.1109/ACCESS.2020.3015243.

[19] T. Kim, **L. F. Vecchietti**, K. Choi, S. Lee, D. Har, "**Machine Learning for Advanced Wireless Sensor Networks: A Review**," in IEEE Sensors Journal, vol. 21, no. 11, pp. 12379-12397, 2021, doi: 10.1109/JSEN.2020.3035846.

[20] S. Kim, I. Kim, **L. F. Vecchietti**, D. Har, "**Pose Estimation Utilizing a Gated Recurrent Unit Network for Visual Localization**", in Applied Sciences, 10, no. 24: 8876, 2020, https://doi.org/10.3390/app10248876.

[21] S. Lee, D. Har, **L. F. Vecchietti**, J. Hong, H. -J. Lim, "**Optimal Link Scheduling Based on Attributes of Nodes in 6TiSCH Wireless Networks**", in The Journal of Korean Institute of Information Technology, vol. 18, no. 1, pp.77-92, 2020, https://doi.org/10.14801/jkiit.2020.18.1.77.

[22] S. Lee, H. Jin, **L. F. Vecchietti**, J. Hong, K-B. Park, PN. Son, D. Har, "**Cooperative decentralized peer-to-peer electricity trading of nanogrid clusters based on predictions of load demand and PV power generation using a gated recurrent unit model**", in IET Renewable Power Generation, vol. 15, pp. 3505-3523, 2021, https://doi.org/10.1049/rpg2.12195.

Preprints

[1] E. Sivak et al. *(Community Paper)*, "**Births are difficult to predict even with rich survey and full-population register data**," ArXiv, 2026.

[2] L. Vecgaile, A. Spata, **L. F. Vecchietti**, E. Zagheni, "**Predicting Individual Life Trajectories: Addressing Uncertainty in Social Employment Transitions**," SocArXiv, 2025.

Workshops, Domestic Conferences

[1] B. N. Wijaya, **L. F. Vecchietti**, A. Cho, C. Cha, M. Cha "**Longitudinal Identification of Critical Factors in Nurse Turnover for Supporting Workforce Well-Being**", in HCI Korea, 2026.

[2] J. Kim, J. Kwon, **L. F. Vecchietti**, A. Oh, M. Cha, "**Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment**," arXiv preprint arXiv:2504.10886 (2025), presented at the ICLR 2025 BiAlign Workshop.

[3] B. N. Wijaya, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Evaluating Antibody Structure Reconstruction with an SE(3)-Equivariant Graph Neural Network**", in Proceedings of the Korea Software Congress (KSC), 2023 *(Excellence Award)*.

[4] M. Lee, **L. F. Vecchietti**, H. Jung, M. Cha, H. M. Kim, "**Protein Sequence Design in a Latent Space via Model-based Reinforcement Learning**", 2022, presented at the NeurIPS 2022 Machine Learning in Structural Biology Workshop.

[5] M. Lee, A. Rzayev, H. Jung, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Structure-based representation for protein functionality prediction using machine learning**", in Proceedings of the Korea Computer Congress (KCC), 2022 *(Excellence Award)*.

[6] P. K. Rajendran, S. Mishra, **L. F. Vecchietti**, D. Har, "**RelMobNet: End-to-end relative camera pose estimation using a robust two-stage training**", 2022, arXiv preprint arXiv:2202.12838, presented at the ECCV 2022 IWDSC Workshop.

[7] B. Hangeldiyev, A. Rzayev, A. Armanuly, **L. F. Vecchietti**, M. Cha, H. M. Kim, "**Antibody Sequence Design With Graph-Based Deep Learning Methods**", in Proceedings of the Korea Software Congress (KSC), 2022.

Work Experiences

Hyundai Motor Company, Namyang Research and Development Center *Namyang, South Korea, Jul. 2013*

Intern. Advisor: Dong-pil Yoon.

Honors and Awards

Science without Borders Program *Feb. 2013 - Feb. 2014*

Brazilian government 1-year scholarship for undergraduate students with outstanding academic achievements.

KAIST Scholarship *Aug. 2017 - Feb. 2021*

Ph.D. Scholarship.

Academic Services

Reviewer

Journals: IEEE Transactions on Cybernetics, IEEE Sensors, IEEE Transactions on Games, Frontiers in AI and Robotics, ACM Transactions on Intelligent Systems and Technology

Conferences: AAAI ICWSM 2022, NeurIPS 2024-2026, ICLR 2025-2027, WSDM 2025, AISTATS 2025, ICML 2025-2026, AAAI 2026, Machine Behavior (Ma+Be) Conference 2026

Workshops: ICML LatinX 2021, ICLR Reincarnating RL 2023, NeurIPS MLSB 2023-2025, ICML ML4LMS 2023, ICML MALGAI 2026

Board Member

Diversity Board - Max Planck Institute for Security and Privacy (MPI-SP) *2025 - Present*

Invited Presentations

Center for Neuroscience-inspired AI - KAIST, South Korea *October 2025*

Invited to present a lecture titled "*From self-organized networks to deep reinforcement learning: perspectives on AI research*."

Ruhr University Bochum, Germany *July 2025*

Invited to present a lecture titled "*LLMs outside Natural Language Processing applications*."

WebImmunization Seminar, University of Oslo, Norway *December 2024*

Invited to present a talk titled "*Integrating Data Science and AI methods in multidisciplinary research to make discoveries with social impact.*"

Cradle Bio, Zurich, Switzerland *November 2024*

Invited to present a talk titled "*Robust Optimization in Protein Fitness Landscapes Using Reinforcement Learning in Latent Space*."

Graduate School of AI, Gwangju Institute of Science and Technology (GIST), South Korea *July 2023*

Invited to present a talk titled "*Developing and applying deep learning methods for protein design*."

Max Planck Institute for Security and Privacy (MPI-SP), Germany *May 2023*

Invited to present a talk titled "*Developing and applying deep learning methods to facilitate new scientific discoveries.*"

IBS Winter School on AI-Boosted Basic Science - Institute for Basic Science (IBS), South Korea *Dec. 2022*

Invited to present a talk titled "*Target-conditioned protein and antibody design for drug discovery.*"

School of AI Convergence - Chonnam National University, South Korea *Nov. 2021*

Invited to present a work titled "*Identifying the key actions that lead an agent to accomplish a task in model-based deep reinforcement learning*."

Cho Chun Shik Graduate School of Green Transportation - KAIST, South Korea *Oct. 2021*

Invited to present a work titled "*Performance enhancement in multigoal model-based deep reinforcement learning*."

Institute for Basic Science (IBS), South Korea *Apr. 2021*

Invited to present a work titled "*Identifying the key actions that lead an agent to accomplish a task in model-based deep reinforcement learning*."

Open-Source Systems

Dino Toothfier (lead developer) [Code]

Machine learning-based application to classify Dinosaur teeth.

AI Soccer Robot Simulator (lead developer) [Code]

Simulator to train multi-agent AI algorithms for robot soccer.